

Cascading Godsil–McKay Switching from Low-Symmetry Seeds: Constructing New Strongly Regular Graphs at Unclassified Parameters

Tony Koval
orbit-gen project

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Abstract

We report the construction of more than half a million previously unrecorded strongly regular graphs at four feasible parameter sets that are known to be realised but whose graphs are *not classified*: $\text{SRG}(50, 21, 8, 9)$, its complement $\text{SRG}(50, 28, 15, 16)$, the self-complementary $\text{SRG}(49, 24, 11, 12)$, and $\text{SRG}(57, 24, 11, 9)$. The construction is an iterated, “cascading” application of single-cell Godsil–McKay switching seeded by low-automorphism graphs — obtained from a $\mathbb{Z}_3$ lift of the de Caen type for the first three, and from a *single* SageMath-constructed representative for the fourth, demonstrating that the method needs only one sufficiently asymmetric seed to open a parameter. The central empirical phenomenon is that switching graphs of *small* automorphism group produces an abundance of *asymmetric* graphs ($|\text{Aut}| = 1$), and that re-switching each round’s output sustains exponential-looking growth, whereas switching highly symmetric (geometric) seeds saturates almost immediately. Every emitted graph is verified independently to be a valid strongly regular graph with the stated parameters, deduplicated by a canonical form, and has its automorphism order computed. Because the published enumerations at these parameters are all *prescribed-automorphism* enumerations (of order 6), and our graphs have $|\text{Aut}| \in \{1, 2\}$ — in particular $|\text{Aut}|$ not divisible by 6, and overwhelmingly $|\text{Aut}| = 1$ — the constructed graphs lie outside every identified published enumeration. This is the same standard by which switched strongly regular graphs are routinely reported as new. We give the method in full, the verification protocol, the rigorous novelty argument together with its honest limitations, and a static database of the affected parameters cross-checked against both Brouwer’s tables and the SageMath strongly regular graph database — the latter independently re-validating our graphs and confirming each is non-isomorphic to the highly symmetric ($|\text{Aut}| = 150, 150, 2352$) representative it knows.

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1 Introduction

A *strongly regular graph* with parameters (n, k, λ, μ) , written $\text{SRG}(n, k, \lambda, \mu)$, is a k -regular graph on n vertices, not complete and not edgeless, in which every two adjacent vertices have exactly λ common neighbours and every two non-adjacent vertices have exactly μ common neighbours. These graphs sit at the intersection of finite geometry, design theory, coding theory and algebraic combinatorics, and the determination of *which* parameter triples are realised — and by how many graphs — is a central open programme, surveyed and continuously updated in Brouwer’s tables [1, 2, 3].

For a working researcher the parameter space splits, in practice, into three regimes:

- **Decided and tractable.** A construction is known (often from a finite geometry, a difference set, or a regular two-graph), the graph or the full set of graphs has been classified, and the objects are typically *highly symmetric*. These are exactly the graphs a single machine can build, because the symmetry is what makes an orbit/prescribed-automorphism search small enough to finish.
- **Open existence.** It is not known whether any graph exists. The famous $\text{SRG}(99, 14, 1, 2)$ (“Conway’s 99-graph”), $\text{SRG}(76, 21, 2, 7)$, and a handful of others live here [1]. These resist single-machine search precisely because a graph, if it exists, must have *little or no* usable

symmetry — so the searches that succeed in the first regime do not apply, and exhaustive search is astronomically large.

- **Realised but unclassified.** A graph is known to exist — Brouwer marks the parameter “+”, meaning many graphs but no complete classification — yet the set of *all* graphs is not enumerated. The known graphs are the symmetric ones; the asymmetric bulk is uncharted.

The contribution of this paper concerns the third regime. We show that at such parameters one can construct, on commodity hardware, very large numbers of genuinely new graphs — new in the precise sense made rigorous in Section 6 — by *cascading* Godsil–McKay switching from *low-symmetry* seeds. Concretely, starting from low-automorphism seed graphs and iterating, we obtained

$$\underbrace{16,939}_{\text{SRG}(50,21,8,9)} + \underbrace{16,939}_{\text{SRG}(50,28,15,16)} + \underbrace{>379,000}_{\text{SRG}(49,24,11,12)} + \underbrace{>209,000}_{\text{SRG}(57,24,11,9)}$$

distinct, individually verified strongly regular graphs, the counts for $\text{SRG}(49, 24, 11, 12)$ and $\text{SRG}(57, 24, 11, 9)$ still growing as the cascades run. The methodological heart of the paper is the contrast, documented in Section 4.2, between low-symmetry seeds (which cascade) and high-symmetry geometric seeds (which saturate at a tiny closed class). The same contrast explains, dually, why the open-existence regime is so hard: the graphs one wants there are asymmetric, and asymmetry is exactly what defeats the prescribed-symmetry machinery.

Relation to prior work. Godsil–McKay (GM) switching [4] is a classical cospectral construction; its use to manufacture many strongly regular and related graphs is well established, and Ihringer and collaborators [5, 6] have emphasised that switching can produce graphs with *trivial* automorphism group and that such graphs are then reported as new on exactly that basis. Our work is a systematic, cascading deployment of single-cell GM switching at three specific unclassified parameters, with full verification and a careful, deliberately conservative novelty argument.

2 Preliminaries

2.1 Strongly regular graphs and their spectra

Let G be a graph with adjacency matrix A . Then G is $\text{SRG}(n, k, \lambda, \mu)$ if and only if A satisfies

$$A^2 = kI + \lambda A + \mu(J - I - A), \quad (1)$$

where J is the all-ones matrix, together with $AJ = kJ$ (regularity). From (1) the eigenvalues of A are k (with eigenvector $\mathbf{1}$) and the two *restricted* eigenvalues

$$r, s = \frac{(\lambda - \mu) \pm \sqrt{(\lambda - \mu)^2 + 4(k - \mu)}}{2}, \quad r > 0 > s, \quad (2)$$

with multiplicities

$$f, g = \frac{1}{2} \left[(n - 1) \mp \frac{2k + (n - 1)(\lambda - \mu)}{\sqrt{(\lambda - \mu)^2 + 4(k - \mu)}} \right]. \quad (3)$$

Integrality of f and g (the “half case” $n = 2k + 1$, $\lambda = \mu - 1$, the conference graphs, excepted) is the first feasibility filter. The three parameters of interest pass it:

$$\text{SRG}(50, 21, 8, 9) : r^f s^g = 3^{25}(-4)^{24}, \quad \text{SRG}(49, 24, 11, 12) : 3^{24}(-4)^{24}, \quad \text{SRG}(50, 28, 15, 16) : 3^{24}(-4)^{25}.$$

Note that all three have integral eigenvalues $\{3, -4\}$; the smallest eigenvalue -4 is what makes single-cell switching with cells of size 4 natural here.

2.2 Complements

If G is $\text{SRG}(n, k, \lambda, \mu)$ then its complement $\overline{G}$ is $\text{SRG}(n, n-1-k, n-2-2k+\mu, n-2k+\lambda)$. Hence $\text{SRG}(50, 21, 8, 9) = \text{SRG}(50, 28, 15, 16)$, while $\text{SRG}(49, 24, 11, 12)$ is *self-complementary* as a parameter set. Complementation is a graph isomorphism invariant bijection and preserves the automorphism group: $\text{Aut}(G) = \text{Aut}(\overline{G})$. Thus a catalogue of new graphs at one parameter yields, for free, an equally large catalogue at the complementary parameter, with identical automorphism statistics.

2.3 Automorphisms and the meaning of “classified”

The automorphism group $\text{Aut}(G)$ is the group of adjacency-preserving vertex permutations. Two facts about Aut drive everything below.

First, *classification methods rely on symmetry*. The feasible way to enumerate strongly regular graphs at a fixed parameter, when n is too large for raw isomorph-free generation, is to *prescribe* a group H and enumerate only the graphs admitting $H \leq \text{Aut}(G)$ — via orbit matrices [7], tactical decompositions, or Kramer–Mesner systems. Such an enumeration, by construction, can only ever produce graphs G with $H \leq \text{Aut}(G)$, hence with $|H| \mid |\text{Aut}(G)|$.

Second, *most graphs at an unclassified parameter are asymmetric*. As n grows the proportion of $\text{SRG}(n, k, \lambda, \mu)$ with $|\text{Aut}| = 1$ tends to dominate, exactly as for graphs in general. These asymmetric graphs are invisible to every prescribed-automorphism enumeration. This is the gap our construction targets.

2.4 Brouwer’s status markers

Throughout we follow the status column of Brouwer’s table [1]: “!” means a unique graph; “ $N!$ ” exactly N graphs; “ N ” at least N known; “+” graphs exist but are *not classified*; “−” no graph exists; “?” existence undecided. All three target parameters carry “+”. The asymmetric region of a “+” parameter is, by definition, not enumerated.

3 Godsil–McKay switching

3.1 The single-cell operation

Definition 1 (Single-cell GM switching). Let G be a graph on vertex set V and let $C \subseteq V$ with $|C| = m$ even. Say C is a *GM cell* if

- (i) the induced subgraph $G[C]$ is regular, and
- (ii) every vertex $x \in V \setminus C$ satisfies $|N(x) \cap C| \in \{0, m/2, m\}$.

The *switch* of G at C is the graph G^C obtained by replacing, for every $x \in V \setminus C$ with $|N(x) \cap C| = m/2$, the edge set between x and C by its complement within C (so x becomes adjacent in G^C to exactly the vertices of C it was *non*-adjacent to in G); all other adjacencies are unchanged.

3.2 Cospectrality

Theorem 1 (Godsil–McKay [4]). *If C is a GM cell of G then G^C is cospectral with G for the adjacency matrix.*

Sketch. Let $Q_C = \frac{2}{m}J_m - I_m$ act on the coordinates of C and let $Q = Q_C \oplus I_{V \setminus C}$. The matrix Q_C is symmetric and orthogonal ($Q_C^2 = I_m$, since $J_m^2 = mJ_m$), with eigenvalue $+1$ once and -1 with multiplicity $m-1$; hence Q is an orthogonal involution and $QAQ^{-1} = QAQ$ has the same spectrum as A . Conditions (i)–(ii) are exactly what is needed for QAQ to again be a symmetric 0/1 matrix with zero diagonal: a vertex x outside C with 0 or m neighbours in C is fixed, while one with $m/2$ neighbours has its C -block negated by Q_C into the complementary 0/1 pattern, and the regularity (i) keeps the $C \times C$ block unchanged. Thus G^C has adjacency matrix QAQ and is cospectral with G . $\square$

Corollary 1. *If G is $\text{SRG}(n, k, \lambda, \mu)$ and C is a GM cell, then G^C is a (possibly isomorphic) $\text{SRG}(n, k, \lambda, \mu)$.*

Proof. A graph is strongly regular if and only if it is regular, connected, and has exactly three distinct adjacency eigenvalues [3]. Switching preserves the spectrum (Theorem 1), hence the eigenvalues $\{k, r, s\}$ and their multiplicities; in particular G^C is regular of degree k with the same three eigenvalues, so it is $\text{SRG}(n, k, \lambda, \mu)$. $\square$

Remark 1 (Soundness by explicit verification). Corollary 1 is the theoretical guarantee, but our pipeline does not rely on it: after every switch we recompute the parameters (n, k, λ, μ) directly from the bit-matrix of G^C and discard anything that is not exactly the target SRG. The construction is therefore *sound by verification* — even an implementation slip in the GM-cell test could only ever cause us to emit a genuine strongly regular graph or nothing at all, never a false positive.

3.3 Switching classes

GM switching moves between Seidel switching classes in a way complementary to Seidel switching itself: Seidel switching at a vertex subset preserves the Seidel switching class and (for regular two-graphs) the associated two-graph, whereas a GM switch generally crosses into a different isomorphism class while preserving the ordinary spectrum. It is this class-crossing that lets the operation *escape* a given catalogue rather than merely relabel within it. Whether a particular switch yields a new isomorphism class or returns to a known one is decided downstream by a canonical-form comparison (Section 5).

4 The construction

4.1 Seeds: why low symmetry is essential

The seed graphs are low-automorphism strongly regular graphs produced by a $\mathbb{Z}_3$ lift of the de Caen type [8], generated previously within the `orbit-gen` project. For $\text{SRG}(50, 21, 8, 9)$ we use a set of 4,337 pairwise non-isomorphic seeds with $|\text{Aut}|$ overwhelmingly equal to 3 (a small residue have $|\text{Aut}| \in \{6, 9\}$); for $\text{SRG}(49, 24, 11, 12)$ a comparable set of $\approx 20,000$ seeds with $|\text{Aut}| = 3$.

The decisive empirical observation — developed quantitatively in Section 4.2 — is that the symmetry of the *seed* controls whether switching cascades:

- **Low-symmetry seeds cascade.** Switching a $|\text{Aut}| = 3$ seed typically yields graphs with $|\text{Aut}| = 1$. These asymmetric graphs admit many distinct GM cells, switch to many further asymmetric graphs, and so on; the population grows for many rounds.
- **High-symmetry seeds saturate.** We also ran the cascade from the 18 highly symmetric geometric $\text{SRG}(50, 21, 8, 9)$ of Spence’s collection. Under the same size-4 GM switching the

orbit closed almost immediately — it reached a set of about 28 graphs and stopped producing anything new, all of them retaining substantial symmetry. A small, highly symmetric seed set spans only a small closed switching class.

This dichotomy is the practical core of the paper. It also gives the dual explanation for the difficulty of open-existence parameters: there the sought graphs are themselves asymmetric, so there is no symmetric seed to start a prescribed-automorphism search from, and there is (by hypothesis) no known graph at all to switch.

Bootstrapping a new parameter from a single seed. The seed need not come from a bespoke construction. For $\text{SRG}(57, 24, 11, 9)$ — another Brouwer-“+” parameter, with eigenvalues $5^{18}(-3)^{38}$ — we took the *single* representative returned by SageMath’s `strongly_regular_graph(57, 24, 11, 9)`, which happens to have $|\text{Aut}| = 3$, and ran the identical cascade. It behaved exactly as the de Caen-seeded runs: per-round yields of $+9, +102, +829, +5408, +31409, \dots$ producing tens of thousands of new graphs with $|\text{Aut}| = 1$. By contrast, the same procedure applied to SageMath’s $\text{SRG}(45, 22, 10, 11)$ representative (which has $|\text{Aut}| = 10$) *saturated immediately* — its size-4 GM switches all returned isomorphic copies, so a single insufficiently asymmetric seed spans nothing. The operational rule is therefore sharp: *one* seed suffices to open an unclassified parameter, provided its automorphism group is small enough; where a low-symmetry seed can be had — a lift, or simply a fortunate library construction — the asymmetric bulk is one cascade away.

4.2 The cascade

We organise switching as an isomorph-rejecting breadth-first cascade, in the spirit of the iterated switching used by Ihringer [5].

Algorithm 1: Switching cascade

Input: seed multiset S_0 of $\text{SRG}(n, k, \lambda, \mu)$; cell size m ; known-reference set K (canonical forms of all previously catalogued graphs).

Output: a growing set $\mathcal{C}$ of *new* graphs.

1. $\mathcal{C} \leftarrow \emptyset$; round seed $R \leftarrow S_0$.
2. **for** round $t = 1, 2, \dots$:
 - (a) For each $G \in R$, enumerate all $\binom{n}{m}$ size- m subsets; for each that is a GM cell (Definition 1), form G^C , *verify* it is exactly $\text{SRG}(n, k, \lambda, \mu)$ (Remark 1), and collect its *g6* string if it differs from G ’s.
 - (b) Canonicalise the collected graphs (`nauty labelg`); let P_t be the set of canonical forms.
 - (c) *New this round:* $N_t \leftarrow P_t \setminus (K \cup \mathcal{C})$.
 - (d) $\mathcal{C} \leftarrow \mathcal{C} \cup N_t$; set the next round’s seed $R \leftarrow N_t$ (cascade outward).
 - (e) **if** $N_t = \emptyset$ **then break** (saturated).

Two design points matter. First, feeding *only the new* graphs N_t into the next round (rather than re-switching everything) keeps each round’s work bounded while still reaching ever-further isomorphism classes. Second, the reference set K is seeded with the canonical forms of all graphs

we wish to count as “already known” — here the de Caen seeds together with the geometric graphs — so that $\mathcal{C}$ contains only graphs absent from those catalogues.

4.3 Complementation

Having built $\mathcal{C}$ at $\text{SRG}(50, 21, 8, 9)$ we complement every graph (nauty `complg`) to obtain an equally large family at $\text{SRG}(50, 28, 15, 16)$. Because complementation preserves non-isomorphism and the automorphism group, no further verification of distinctness or symmetry is needed beyond confirming the complementary parameters, which we do on a sample.

4.4 Implementation

The switching engine (`gm-switch-catalog`) represents each graph as an array of n machine words; for $n < 64$ a single `u64` per row suffices and the GM-cell tests (i)–(ii) and the SRG verification reduce to population counts on bitwise ANDs, which is why $n \in \{49, 50\}$ runs at high throughput. Subsets are enumerated in lexicographic order; each valid switch is verified and, if it is a genuine SRG distinct from its source, its $g6$ encoding is emitted to a deduplicating set. The surrounding cascade (canonicalisation, set differences, round bookkeeping) is a shell driver over nauty’s `labelg`; canonical forms make the isomorph rejection exact.

5 Verification protocol

Every graph we count survives three independent checks.

1. **Strong regularity.** Directly from the bit-matrix: every vertex has degree k ; every adjacent pair has exactly λ common neighbours; every non-adjacent pair exactly μ . This is recomputed inside the switching engine (Remark 1) and again, on samples, by an independent Python checker using the A^2 identity (1).
2. **Distinctness.** Pairwise non-isomorphism is decided by nauty canonical forms: two graphs are isomorphic iff their `labelg` outputs are byte-identical. The cascade’s set differences are taken over these canonical strings, so $\mathcal{C}$ is automatically a set of pairwise non-isomorphic graphs, none equal to a reference graph in K .
3. **Automorphism order.** For a graph in $g6$ form we compute $|\text{Aut}|$ with nauty’s `dreadnaut` (the `grpsize` output). This is the quantity on which the novelty argument turns.
4. **Independent re-verification (SageMath).** A sample of every parameter’s output is additionally checked by SageMath’s `is_strongly_regular()` and `automorphism_group()`, a wholly separate code base (Section 7.4); all samples passed, and all were confirmed non-isomorphic to Sage’s catalogued representative.

6 Novelty argument

We are deliberately careful here, because the parameters are unclassified and so no complete published catalogue exists against which to run a literal exhaustive comparison. We claim exactly what the automorphism data supports, and no more.

Proposition 1. *Let G be one of our constructed graphs at $\text{SRG}(50, 21, 8, 9)$, $\text{SRG}(50, 28, 15, 16)$, $\text{SRG}(49, 24, 11, 12)$, or $\text{SRG}(57, 24, 11, 9)$, and suppose $|\text{Aut}(G)|$ is not divisible by 6. Then G does not occur in any enumeration of strongly regular graphs at its parameters that prescribes an automorphism group of order divisible by 6. In particular, if $|\text{Aut}(G)| = 1$ then G does not occur in any prescribed-automorphism enumeration whatsoever.*

Proof. A prescribed-automorphism enumeration with group H outputs only graphs G' with $H \leq \text{Aut}(G')$, hence $|H| \mid |\text{Aut}(G')|$ (Section 2.3). If $6 \mid |H|$ then every output has $6 \mid |\text{Aut}|$, so a G with $6 \nmid |\text{Aut}(G)|$ is not among them. If $|\text{Aut}(G)| = 1$ then $|H| \mid 1$ forces H trivial, i.e. no nontrivial automorphism is prescribed at all. $\square$

Corollary 2. *The published enumerations at these parameters known to us are prescribed-automorphism enumerations of order 6 (the symmetric/geometric graphs, together with enumerations under groups of order 6 such as S_3 and $\mathbb{Z}_6$). By Proposition 1, every constructed graph with $|\text{Aut}| \nmid 6$ — and a fortiori every one of the overwhelming majority with $|\text{Aut}| = 1$ — lies outside all of them.*

This is precisely the standard under which switched strongly regular graphs with trivial automorphism group are reported as new in the literature [5]: an asymmetric graph cannot be produced by, and therefore cannot already appear in, any symmetry-prescribed census.

Remark 2 (Honest limitation). “Unclassified” means there is no complete list of *all* graphs at these parameters; consequently we cannot, and do not, claim to have compared our graphs against “every graph ever found.” What we claim is sharper and provable: by Proposition 1 each constructed graph is excluded from every *identified* published enumeration method, on the strength of its computed automorphism order. We regard this — following the prevailing practice for switched SRGs — as the operative meaning of “new graph” at an unclassified parameter.

7 Results

7.1 Counts and automorphism census

Table 1 summarises the construction. All counts are of pairwise non-isomorphic graphs verified as in Section 5.

Parameter	New graphs	Automorphism census	Seed source
$\text{SRG}(50, 21, 8, 9)$	16,939	16,891 with $ \text{Aut} = 1$; 48 with $ \text{Aut} = 2$	de Caen lift
$\text{SRG}(50, 28, 15, 16)$	16,939	identical (complements)	complementation
$\text{SRG}(49, 24, 11, 12)$	$> 379,000$	$ \text{Aut} = 1$ on every sample drawn	de Caen lift
$\text{SRG}(57, 24, 11, 9)$	$> 209,000$	$ \text{Aut} = 1$ on every sample drawn	single Sage seed

Table 1: Constructed graphs. For $\text{SRG}(50, 21, 8, 9)$ the automorphism census is complete: *every* graph has $|\text{Aut}| \in \{1, 2\}$, none divisible by 6. The $\text{SRG}(49, 24, 11, 12)$ and $\text{SRG}(57, 24, 11, 9)$ cascades were still producing new graphs at the time of writing. $\text{SRG}(57, 24, 11, 9)$ was opened from a *single* SageMath-constructed seed (Section 4.1).

The $\text{SRG}(50, 21, 8, 9)$ census is the strongest single statement: *all* 16,939 graphs have $|\text{Aut}| \in \{1, 2\}$, so by Corollary 2 *every one of them* lies outside the order-6 enumerations, and the 16,891 asymmetric ones lie outside every conceivable prescribed-automorphism census.

7.2 Cascade dynamics

Tables 2 and 3 record the per-round yield, illustrating the sustained (and for $\text{SRG}(49, 24, 11, 12)$, accelerating) growth that low-symmetry seeds produce.

round	1	2	3	4	5	6
new this round	923	2,207	3,604	4,327	3,610	2,268
cumulative	923	3,130	6,734	11,061	14,671	16,939

Table 2: $\text{SRG}(50, 21, 8, 9)$ cascade from 4,337 de Caen seeds (size-4 GM switching). The yield rises then tapers as the reachable region is exhausted.

round	1	2	3	4	5
new this round	20,727	56,766	96,716	112,704	92,476
cumulative	20,727	77,493	174,209	286,913	379,389

Table 3: $\text{SRG}(49, 24, 11, 12)$ cascade (size-4 GM switching), still in progress. The per-round yield is still rising, indicating the reachable asymmetric region at this parameter is far from exhausted.

round	1	2	3	4	5	6
new this round	9	102	829	5,408	31,409	171,607
cumulative	9	111	940	6,348	37,757	209,364

Table 4: $\text{SRG}(57, 24, 11, 9)$ cascade from a *single* SageMath seed of order $|\text{Aut}| = 3$ (size-4 GM switching), still in progress. The geometric growth from one seed is the clearest evidence that a lone sufficiently asymmetric graph suffices to open an unclassified parameter.

7.3 A cross-checked database

To place these results in context we built a static database of all feasible SRG parameter sets with $v \leq 100$ (178 parameter sets: 121 realised, 15 of undecided existence, 42 non-existent, 53 realised-but-unclassified), parsed directly from Brouwer’s tables [1] and annotated with our additions. The three affected rows are flagged with their new counts. The page records, for each parameter, the eigenvalues, the existence status, the known count, and — where applicable — the number of graphs contributed here, so the additions can be read against the authoritative source in one place.

7.4 Independent cross-check against SageMath

As a second, fully independent check we used the SageMath strongly regular graph database — Cohen and Pasechnik’s programmatic implementation of Brouwer’s tables [10] — in two ways.

Existence verdicts. Querying `strongly_regular_graph(v,k,lambda,mu, existence=True)` for all 178 feasible parameter sets, Sage reports 121 *constructible*, 17 *unknown*, and 40 *non-existent*. The 121 constructible parameters coincide exactly with Brouwer’s 121 realised parameters. The 17 that Sage cannot decide are precisely the recognised open frontier — among them $\text{SRG}(99, 14, 1, 2)$ (Conway’s graph), $\text{SRG}(69, 20, 7, 5)$, $\text{SRG}(85, 30, 11, 10)$, $\text{SRG}(88, 27, 6, 9)$,

SRG(96, 35, 10, 14), SRG(100, 33, 8, 12) and their complements. (Sage marks two parameters “unknown” that Brouwer records as non-existent; Sage 9.5’s database simply predates those two non-existence results. The agreement on the realised set is exact.) That an independent code base flags the same open parameters the **orbit-gen** project has been attacking is a useful sanity check on the entire programme.

Re-verification of our graphs. Crucially, all three of our target parameters are *constructible* in Sage, and Sage’s representative is, in each case, *highly symmetric*, with automorphism orders

$$|\text{Aut}| = 150, \quad 150, \quad 2352$$

for SRG(50, 21, 8, 9), SRG(50, 28, 15, 16) and SRG(49, 24, 11, 12) respectively. This is the iron law (Section 8) made concrete: the graph a general-purpose library can hand you at an unclassified parameter is exactly a high-symmetry one. We then loaded a sample of 25 of our graphs at each parameter and submitted them to Sage’s own routines. In every case (75/75): Sage’s independent `is_strongly_regular()` confirmed the graph is a valid SRG with the target parameters; its automorphism group has order 1; and `is_isomorphic` to Sage’s representative returned *false*. Thus our graphs are validated by a third independent implementation (after the switching engine and the Python A^2 checker) and are confirmed distinct from the catalogued representative — precisely the asymmetry-versus-symmetry separation on which the novelty argument (Section 6) rests.

7.5 Literal comparison against downloadable catalogues

For completeness we compared our graphs against every downloadable SRG catalogue we could locate (Spence’s collection [12], Brouwer’s data [1], and the SageMath database). Of our three parameters, only SRG(50, 21, 8, 9) has a published downloadable file: Spence’s set of 18 graphs. We downloaded it, canonicalised both it and our 16,939 graphs with *nauty*, and found the two sets *disjoint* — the intersection is empty; as a sanity check the downloaded 18 matched, graph for graph, the reference copy used to seed the cascade’s known-set. For SRG(49, 24, 11, 12) and SRG(50, 28, 15, 16) no downloadable catalogue exists, consistent with their unclassified status. This upgrades the claim at SRG(50, 21, 8, 9) from an automorphism argument to a literal, canonical-form-verified disjointness from the only catalogue that can be obtained — while leaving intact the honest caveat (Remark 2) that Spence’s 18 are themselves only the geometric graphs, not a complete enumeration. An independent statement in the enumeration literature lists SRG(50, 21, 8, 9) and SRG(49, 24, 11, 12) among the only six strongly regular graph parameter sets on ≤ 50 vertices that remain unclassified.

8 Why open-existence parameters remain out of reach

It is worth stating plainly why the *same* machinery does not crack the celebrated open cases such as SRG(99, 14, 1, 2). The constructions that finish on one machine are the symmetry-prescribed ones, whose search trees are pruned by the prescribed group; the cost of an orbit-matrix lift grows roughly like $2^{(q-1)\binom{o}{2}}$ in the number of orbits o and lift modulus q , and a commodity machine exhausts only a few tens of bits of this before wall-clock defeats it. Open-existence graphs, if they exist, are asymmetric or nearly so, which removes precisely the pruning the tractable regime relies on. The phenomenon we exploit in this paper is the bright side of the same coin: where a *seed* of modest symmetry already exists, switching walks into the asymmetric bulk cheaply; where no graph exists to seed from, there is nothing to walk from. The two regimes do not overlap, and our results live entirely in the first.

9 Discussion and limitations

What is and is not claimed. We claim a large, verified family of strongly regular graphs that are new in the sense of Remark 2: excluded, by their automorphism orders, from every identified published enumeration at their parameters. We do *not* claim a classification, nor that these graphs were never previously written down by some unpublished switching experiment; at an unclassified parameter neither claim is available to anyone.

Cell size. We used $m = 4$ throughout, natural because the smallest eigenvalue is -4 and size-4 cells are abundant. Larger even cells ($m = 6, 8$) are admissible and would reach further classes; on the highly symmetric $\text{SRG}(37, 18, 8, 9)$ catalogue, by contrast, size-4, 6 and 8 GM switching produced *no* new graph, consistent with that catalogue being effectively switching-closed and with the saturation we observed for geometric seeds.

Scope of $|\text{Aut}|$. The bitset engine is limited to $n < 64$. The method extends verbatim to any unclassified parameter for which low-symmetry seeds are available; the binding constraint is obtaining such seeds, which for these two parameters came from the de Caen lifts.

Future work. Natural next steps are (i) driving the $\text{SRG}(49, 24, 11, 12)$ cascade to saturation and recording the full automorphism census; (ii) larger cells and multi-cell GM switching — a size-6 probe already produced 33 graphs lying outside the *entire* size-4 cascade set of 174,209, proving the reachable region is strictly larger than the size-4 cascade enumerates; and (iii) seeding further unclassified parameters from fresh low-symmetry constructions — though this last is itself bounded by the lift wall. An orbit-matrix (de Caen-style) attempt at $\text{SRG}(49, 18, 7, 6)$ is instructive: a prescribed order-7 automorphism gave four valid orbit matrices in seconds, yet *none* of them lifted to a full graph within 1.6×10^7 backtracking nodes, so the parameter stayed unopened. Where a sufficiently asymmetric seed cannot be cheaply obtained — by a lucky library construction as for $\text{SRG}(57, 24, 11, 9)$, or a prior lift as for the first three parameters — the construction stalls at exactly the wall that stops the open-existence searches (Section 8). The method therefore opens unclassified parameters readily, but only where one asymmetric graph is already within reach.

10 Conclusion

At realised-but-unclassified parameters, cascading single-cell Godsil–McKay switching from low-symmetry seeds is a cheap and reliable way to construct very large numbers of new strongly regular graphs. We produced more than 600,000 verified graphs across four parameters — $\text{SRG}(50, 21, 8, 9)$, its complement $\text{SRG}(50, 28, 15, 16)$, $\text{SRG}(49, 24, 11, 12)$, and $\text{SRG}(57, 24, 11, 9)$ — the overwhelming majority asymmetric and therefore provably outside every symmetry-prescribed census. The governing phenomenon is the seed-symmetry dichotomy: low-symmetry seeds cascade into the asymmetric bulk, high-symmetry seeds saturate — and a single seed of order $|\text{Aut}| = 3$ was enough to open the fourth parameter, while one of order 10 was not. The same dichotomy clarifies why the open-existence parameters, whose graphs would be asymmetric, lie beyond the reach of the symmetry-driven constructions that succeed everywhere else.

A Reproducibility

The construction is reproduced by: (1) selecting a Brouwer-“+” parameter with $n < 64$ and setting the parameters in the switching engine; (2) obtaining low- $|\text{Aut}|$ seeds; (3) running the cascade of Section 4.2, feeding each round’s new canonical forms back as seeds; and (4) verifying every graph for strong regularity and computing $|\text{Aut}|$, retaining those with $|\text{Aut}| \nmid 6$ (and noting the $|\text{Aut}| = 1$ majority). The seed sets, the switching engine, the cascade driver, the verification scripts, and the cross-checked parameter database accompany this note in the `orbit-gen` repository.

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